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# COMPARISON STUDY OF LEFT VENTRICULAR STROKE VOLUME ASSESSMENT USING TWO-DIMENSIONAL AND THREE-DIMENSIONAL COLOR DOPPLER ECHOCARDIOGRAPHY

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## Abstract :

Whether size of left ventricular outflow tract diameter (LVOTd) the usage of colour Doppler (CD) in an effort to extra correctly define LVOTd is more correct for dedication of stroke extent (SV) than gray scale and compare it with direct dimension of LVOT vicinity (a) using 3-dimensional echocardiography (3DE) for SV determination. techniques and consequences: Twenty-one volunteers have been tested. LVOTa was calculated by -dimensional echocardiography (2DE) the usage of the following formula: π× (d/2)2, d = LVOT diameter by grey scale and CD, respectively. Planimetry of LVOTa become performed in parasternal long axis the usage of 3DE. Eccentricity Index changed into calculated using the lateral and anterior-posterior LVOTd. SV changed into received by way of four different methods: second grey scale, 2d coloration, LVOTa × LVOT speed time imperative, and SV with the aid of Simpson's biplane approach. gray scale LVOTd become substantially smaller compared to LVOTd acquired with CD ($P < 0.05$). widespread variations happened among LVOTa gray scale and CD (3.29 ± 0.74 cm2 vs 3.67 ± 0.70 cm2, $P < 0.05$) and among LVOTa calculated by means of grey scale in contrast to 3DE planimetry; (3.29 ± 0.74 cm2 vs 3.61 ± 0.89 cm2, $P = 0.011$). 1/2 of the topics had at the least 17% distinction between the lateral and anterior-posterior LVOTd. There have been sizeable variations between SV via 2d gray scale and 2nd CD (82.8 ± 17.1 mL vs ninety 2.4 ± 16.8 mL, $P < 0.05$) and between 2D gray scale and 3DE planimetry (82.8 ± 17.1 mL vs 90.7 ± 19.8 mL, $P = 0.025$). Conclusion: Our study demonstrates LVOT being frequently elliptical. SV and LVOTa were found to be similar when comparing 2DE CD and 3DE planimetry and showed higher values in comparison to 2DE gray scale, which suggests 2DE CD to be an alternative approach for SV assessment.

**Keywords :** Left ventricular, Left ventricular stroke volume, Color Doppler

## Introduction :

Stroke volume (SV) is an important hemodynamic parameter in clinical medicine providing a guide to the circulatory conditions as well as an evaluation of the response to therapeutic interventions. Two-dimensional echocardiography (2DE) is currently the most widely used noninvasive imaging tool for the anatomical and functional analysis of the heart including the assessment of aortic valve area (AVA) and SV 1-4 Several noninvasive 2DE methods can be used for SV calculation.

The difference between enddiastolic (ED) and end-systolic chamber volumes or left ventricular outflow tract (LVOT) pulsed wave (PW) Doppler velocity time integral (VTI) multiplied by LVOT area (a) for online or offline measurements are two of them. Volume calculations using the Simpson's biplane formula by subtracting left ventricular (LV) ED volume (V) and LVESV for SV assessment are prone to error, since there might be difficulties in defining LV borders or abnormal ventricular geometry, foreshortening due to difficulties acquiring images from the true apex [1].

The SV calculation from the Simpson's biplane rule could lead to inaccurate results if there is a mitral insufficiency. SV determination from LVOT diameter (d) makes several statements, such as (1) the outflow tract has a circular configuration and (2) its dimension remains stable during systole.5 The major difficulty in assessment of SV by calculating LVOTd is that the diameter is squared in order to determine SV and thus small errors in measuring LVOTa are amplified Three-dimensional echocardiography (3DE) assessment of the LV function has proven to be highly accurate when compared with magnetic resonance imaging in different patient categories. This approach might be able to eliminate a serious limitation for the calculation of LVOTa as it permits the direct measurement and assessment of the anatomic reality of this cardiac region. 3DE has also proved to be superior to 2DE methods for chamber volume calculation. We tested the hypothesis that measurement of LVOTd on color Doppler (CD) superimposed images might be more accurate for determination of SV compared to gray scale determination of LVOTd. These measurements were compared to direct LVOTa planimetry by 3DE for SV assessment which was used as reference method.

## Materials and methods:

## Patient Selection:

Twenty-one volunteers (8 men and 13 women, aged 24 ± 5 years) with good to excellent acoustic window and no known cardiac disease were examined. Exclusion criteria were patients with atrial fibrillation/or flutter, poor image quality, and abnormal valvular function (regurgitation). All the participants signed an informed consent allowing the procedures. Before each examination, age, height, weight, and blood pressure were registered. Cardiovascular risk factors were collected with a questionnaire. Table I shows baseline characteristics of the included volunteers. The study was approved by the local ethical committee, Stockholm, Sweden [2].

## Imaging Acquisition:

A complete 2DE and Doppler study was performed in all subjects, using a commercially available Vivid E9 (GE Healthcare, Horten, Norway) ultrasound machine equipped with M5S transducer. All acquisitions were performed by the same operator with the subject in the left lateral position.

VTI of the subvalvular flow was recorded from an apical five-chamber view by placing the sample volume (5 mm) of PW Doppler at the base of the aortic leaflets and then moving slowly away toward LVOT until a typical subvalvular flow profile was seen (0.5 mm on the LV side of the aortic valve). An optimal signal shows a smooth velocity curve with a narrow velocity range at each time point. LVOT zoomed image with and without CD were acquired. The CD (mean aliasing velocity of 60 ± 5 cm/sec) with default settings with a minimum operator adjustment was used in order to achieve standardization of image quality to better define and measure LVOTd in order to calculate SV. Apical two-chamber and four-chamber view were recorded for assessment of SV by Simpson's biplane method.

A novel 4V matrix-array transducer was used for the acquisition of the 3DE data sets from four consecutive cardiac cycles during end expiration breath-hold in parasternal long-axis view (PLAX) and in apical view. This transducer allows the use of the "full-volume" technique with a mean frame rate of 40 ± SD volumes/sec.

All the above-described images were acquired twice and best image quality was chosen for further analysis. Data sets were stored for offline analysis using commercially available software (EchoPAC PC version 110.0.0, GE Healthcare) [3].

## Image Analysis:

The image analysis involved the measurement of several parameters which were done according to the American Society of Echocardiography guidelines. Each parameter was measured twice by two independent readers with 1 week apart blinded to the previous reading. Each observer made three consecutive measurements and the mean value was recorded.

In PLAX, gray scale and color LVOTd were measured 5 mm above the aortic valve in mid systole, according to trailing edge to leading edge5. VTI was measured by tracing the spectral PW Doppler velocity signal using a minimum gain setting and the sweep velocity was set >100 mm/sec for you to attain a higher delineation. This parameter and the approximated go-sectional region of the LVOT have been used for similarly calculation of SV by means of following equation: SV = π × (d/2)2 × LVOTVTI. guide tracing of endocardial border the use of Simpson's biplane method became performed to obtain ejection fraction [1].

After editing the transverse plane orientation of PLAX view acquired by means of 3DE, an idealized LVOT brief-axis view (SAX) turned into carried out and LVOTa became measured 5 mm above the aortic valve by using manually tracing the define of LVOT.

LVOT lateral and anteriorposterior diameters had been additionally measured. Anteriorposterior LVOTd described because the diameter parallel (90 ◦ ) to left atrium (los angeles) and the lateral LVOTd perpendicular to l. a. became calculated.

Those diameters were used to calculate Eccentricity Index (EI)4, EI = 1-(anterior-posterior LVOTd)/(lat LVOTd). An index of zero could represent a circular LVOT, where a better EI describes a noncircular geometry [4].

## Statistical analysis:

All variables have been tested for normality the usage of the Shapiro-Wilk take a look at. Values of $P < 0.05$ were taken into consideration statistically giant. the Bland-Altman evaluation turned into accomplished to determine the systemic bias and limits of settlement of all measurements with the aid of every approach. Group comparison between non-stop variables was done the usage of ANOVA [3].

Paired pattern t-take a look at become executed to examine the importance of distinction [LVOTd gray scale-LVOTd color] variable. Wilcoxon take a look at changed into accomplished to evaluate the lateral and the anteroposterior LVOTd [5].

## Results:

## Intra and Interobserver Variability:

There was good agreement inside (intraobserver variability) and between (interobserver variability) the 2 impartial observers' measurements of all the studied parameters. LVOTVTI became the parameter with least intraobserver variability (1.3%) while the very best variability became determined for LVOTa via 3DE planimetry (5%) (table 1).

Interobserver variability ranged from 2.1% (LVOTd grey scale) to three.6% (LVOTa) (table 1 and 2).

Patient Characteristics (n = 21)

| Variables | *Mean ± SD* |
|---|---|
| Age (yrs) | 24 ± 5 |
| Males (%) | 38 |
| Females (%) | 62 |
| Height (cm) | 171.4 ± 8.9 |
| Weight (kg) | 70.1 ± 12.0 |
| BSA ($m^2$) | 1.8 ± 0.19 |
| SBP (mmHg) | 115 ± 13 |
| DBP (mmHg) | 78 ± 10 |
| Resting HR (bpm) | 59 ± 7 |
| EF by RT3DE (%) | 59 ± 2 |

BSA = body surface area; SBP = systolic blood pressure; DBP = diastolic blood pressure; HR = heart rate; RT3DE = real time three-dimensional echocardiography.

**Table 1**: Patient Characteristics (n = 21).

Different Variables and Their Intra- and Interobserver Variability

| | 2DE LVOTd Gray Scale | LVOTd Color | LVOT VTI | |
|---|---|---|---|---|
| Intra 1 | 1.4% | 1.4% | 1.3% | |
| Intra 2 | 1.5% | 1.7% | 1.7% | |
| Inter 1–2 | 2.1% | 2.5% | 2.5% | |
| | RT3DE LVOT Area | LVOTd Lateral | LVOTd Antero-Posterior | SV |
| Intra 1 | 5.0% | 3.2% | 4.0% | 2.9% |
| Intra 2 | 1.7% | 2.3% | 2.1% | 3.9% |
| Inter 1–2 | 3.6% | 3.0% | 3.3% | 2.8% |

2DE = two-dimensional echocardiography; d = dimension; LVOT = left ventricle outflow tract; VTI = velocity time integral; SV = stroke volume.

**Table 2:** Different Variables and Their Intra- and Interobserver Variability.

## Contrast of LVOT regions:

Mean LVOTa calculated with the aid of grey scale, CD, and 3DE planimetry and the evaluation of suggest SV between the four exclusive techniques are provided. Good sized differences had been located between LVOTa grey scale and CD ($P < 0.05$) and also between LVOTa gray scale and 3DE planimetry ($P = 0.011$). The difference among CD and 3DE planimetry did now not attain the level of statistical importance (table 3) [6,1].

Mean Values of LVOTa and SV Obtained from 2DE and 3DE

| | Mean Difference ± SD | |
|---|---|---|
| | LVOTa ($cm^2$) | SV (mL) |
| 2D Gray scale | 3.29 ± 0.74 | 82.8 ± 17.1 |
| 2D color Doppler | 3.67 ± 0.70 | 92.4 ± 16.8 |
| RT3DE planimetry | 3.61 ± 0.89 | 90.7 ± 19.8 |
| 2D Simpson's method | | 79.3 ± 15.2 |

**Table 3**: Mean Values of LVOTa and SV Obtained.

## LVOT Eccentricity:

The mean fee of the lateral and the anteriorposterior LVOTd diameter was 22.8 ± 2.7 mm and 18.8 ± 2.5 mm, respectively. widespread distinction befell between those diameters with suggest distinction fee of four.05 ± 1.64 mm (P < 0.05). The suggest EI became 0.21 and median of 0.17 (table 4).

Mean Difference and Limits of Agreement of SV Assessed by 2DE and 3DE

| | Mean + SD (mL) | Limits of Agreement | | P-Value |
|---|---|---|---|---|
| 2D gray scale vs 2D color | −9.7 ± 4.9 | −19.5 | 0.2 | 0.001 |
| 2D gray scale vs. RT3DE plan | −7.9 ± 10.5 | −29.0 | 13.1 | 0.025 |
| 2D gray scale vs. Simpson's method | 3.5 ± 10.9 | −18.2 | 25.3 | 1.000 |
| 2D color vs. RT3DE plan | 1.7 ± 12.2 | −22.8 | 26.2 | 1.000 |
| 2D color vs. Simpson's method | 13.2 ± 11.3 | −9.4 | 35.7 | 0.001 |
| RT3DE plan vs. Simpson's method | 11.4 ± 13.7 | −15.9 | 38.8 | 0.010 |

2D = two-dimensional; RT3DE = real time three-dimensional echocardiography; plan = planimetry.

**Table 4:** Mean Difference and Limits of Agreement of SV Assessed by 2DE and 3DE.

## Discussion:

Present day tips by using the yank Society of Echocardiography propose that that allows you to calculate AVA with the aid of continuity equation (CE) both the aortic annular or LVOT diameters have to be used. on account that LVOT may also have an elliptical form, the LVOTd measured in 2DE grey scale is regularly smaller, resulting in great underestimation of the authentic go-sectional region The consequences of this study show that LVOT is not a round shape which is consistent with many other investigations [7]. EI was 0.21 ± 0.13 and its median 0.17, this means that that 1/2 of the subjects had at least 17% distinction among the lateral and the anterior-posterior LVOTd. comparable consequences had been founded by Doddamani et al. that confirmed a mean and median EI of 0.21 and 0.20, respectively [2].

Direct planimetry the usage of 3DE is an increasing number of popular as this method allows real LVOT vicinity to be measured without relying on the belief of circularity and avoids blunders brought about by way of squaring the measured LVOT radius Our group has formerly showed that direct planimetry of LVOT area is an correct and viable method and correlates extraordinarily properly with real time three-D echocardiography (RT3DE) full extent measurements for SV evaluation validated that both multislice computed tomography and 3-d transesophageal echocardiography confirmed an elliptical aortic annular/LVOT geometry, main to good sized underestimations in their "proper" go-sectional areas when assuming a circular geometry.

9 according to our outcomes there are vast variations between LVOTa measured with the aid of 2DE grey scale and 3DE planimetry (P = 0.011) [3]. no matter the reality that 2DE CD entails geometric assumptions, there have been no vast differences observed between this technique and 3DE planimetry. despite the fact that no statistical significance was found, the limits of agreement have been alternatively huge (-0.96 to at least one.07 cm2) which can be of interest within the clinical setting. though our effects recommend that 2DE CD is in higher settlement with 3DE planimetry compared to the corresponding of 2DE gray scale in LVOTd assessment.

In step with this locating another query might be raised, if it's far viable that LVOTd assessed by way of 2DE CD overestimate the diameter. in that case the calculated region by means of CE using this parameter is much like direct planimetry in elliptical LVOT but should lead in overestimation in circular LVOT.

Dedication of LV volumes and EF from 2DE pics may be enormously misguided because of foreshortened views and the usage of geometric as sumptions. Regarding SV size by means of the distinct strategies, higher values have been received with 3DE LVOT planimetry and 2DE CD. In reality, second grey scale resulted in a systematic underestimation of SV in comparison with the corresponding 3D and 2nd CD measurements [4].

This isn't unexpected thinking about the truth that 2nd grey scale is based totally on geometric assumptions of LVOT that do not always correspond to LVOT anatomy. however, colour guidance provide better delineation of the LVOT and yields always large LVOT diameter in better agreement with measurements finished with direct planimetric 3-d evaluation [8].

As supplied in desk V, now not handiest the mean difference between 3D planimetry and 2d CD measurements become decrease than whilst those measurements had been compared to second but the limits of settlement had been narrower as nicely. Additionally, there was no tremendous differences between three-D planimetry and 2nd CD (P = 1.000).

However, in spite of this, concerning SV measurements the limits of agreement had been rather wide among measurements acquired by those novel tactics, i.e., 3D and shade superimposition on 2nd pics, which should be taken into consideration in scientific praxis in particular in sufferers with low output states [1].

In a previous have a look at it was proven that SV measurements with 3-d planimetry have been in appropriate correlation with complete three-D extent measurements and the bounds of agreement between those two techniques have been narrow. aid those findings demonstrating the direct dimension of the LVOTa from 3DE provides correct evaluation of SV. but, SV measurements using 3DE require understanding, luxurious equipments, and devoted software program which may also restriction its use in scientific exercise. taken into consideration this, LVOTd assessed by 2DE CD is simpler to carry out in ordinary medical practice in maximum echocardiography laboratories and may be taken into consideration as an alternative technique for hemodynamic measurements of cardiac output and SV [9].

Further this have a look at was done in sufferers with structurally regular aortic valves. In medical exercise, pathological findings as valvular calcification, with consequent LVOT shadowing or interventricular basal septal hypertrophy, could result in difficulty with the intention to degree LVOTd.16 In these instances the usage of CD may enhance the LVOTd assessment given that it may provide higher delineation of LVOT borders [7].

That is mainly vital given the current hobby in situations which include paradoxical low-go with the flow, low-gradient excessive aortic stenosis.16 destiny studies are needed which will elucidate give up validate using the technique described in this file in this affected person class.

## Conclusion:

Our study showed that LVOT shape is usually now not spherical. Incorrectly assuming a round LVOT geometry using the CE based totally on 2DE grey scale can underestimate LVOTa and consequently SV.

Our effects propose that 2DE CD is similar to 3DE planimetry regarding LVOTa and SV calculation than 2DE grey scale. the usage of 2nd CD in LVOTd size in order to calculate SV and AVA can be taken into consideration as an option in daily exercise, every time the usage of direct planimetry isn't viable.

## Conflict of interest:

Author declares there is no conflict of interest.

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